

Performance Testing

This document records the functional testing, model evaluation, and performance benchmarking results for OptiCrop v1.0.

1. Model Accuracy Visual Comparison

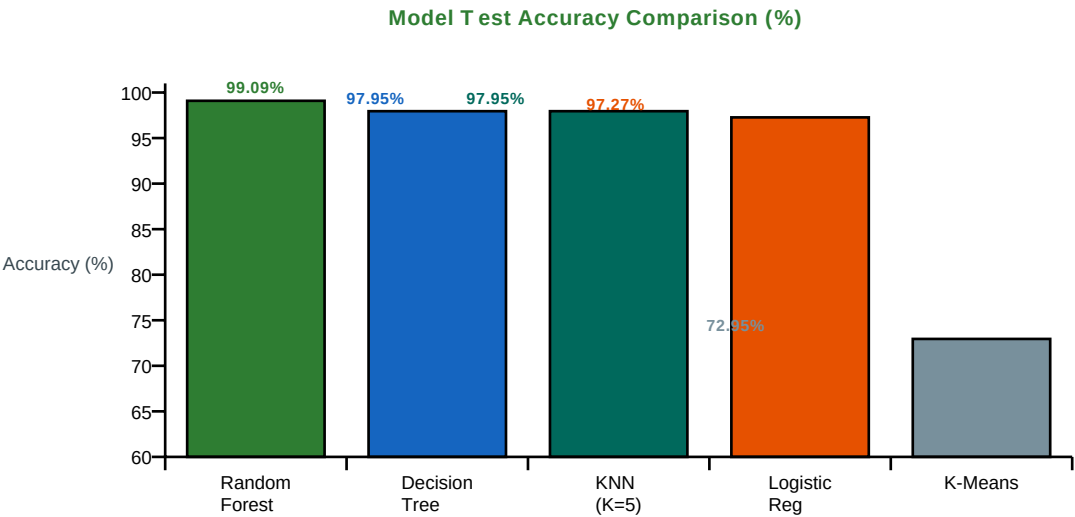


Figure 7: Test Accuracy Comparison Across All 5 ML Algorithms

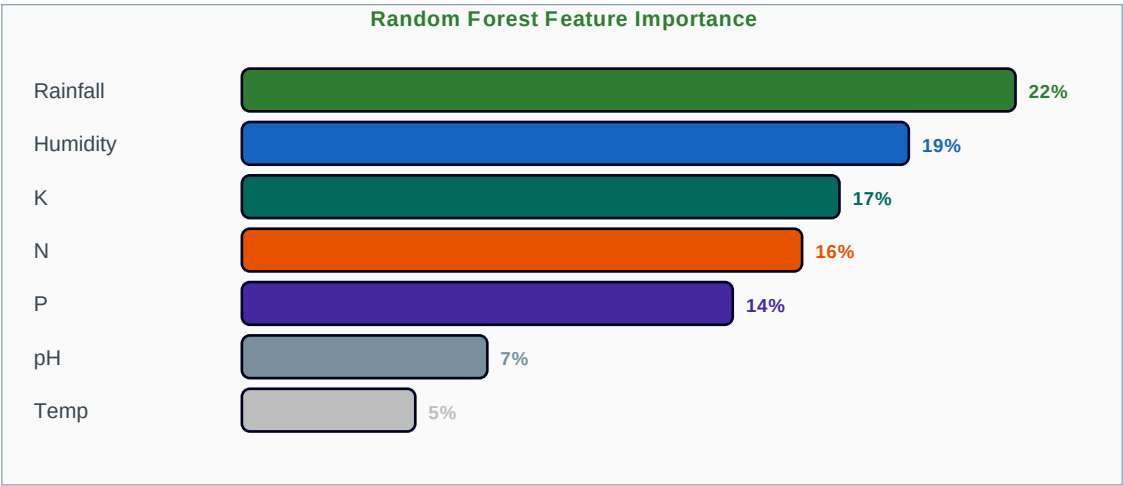


Figure 8: Random Forest Feature Importance (importance score)

2. Model Accuracy Metrics Table

Algorithm	T rain Accuracy	T est Accuracy	CV (5-fold)	Selected
Random Forest (n=100)	100.0%	99.09%	99.0% – 0.4%	YES
Decision Tree	100.0%	97.95%	97.8% – 0.6%	No

K-Nearest Neighbours (K=5)	98.2%	97.95%	97.6% – 0.7%	No
Logistic Regression	96.8%	97.27%	96.9% – 0.8%	No
K-Means Clustering		72.95%		No (unsupervised)

3. Unit Test Results (pytest)

Test ID	Module	Test Description	Result
T01	preprocess.py	validate_inputs() valid rice inputs accepted	PASS
T02	preprocess.py	validate_inputs() N out of range raises ValueError	PASS
T03	preprocess.py	validate_inputs() pH below 3.5 raises ValueError	PASS
T04	preprocess.py	validate_inputs() non-numeric string raises ValueError	PASS
T05	preprocess.py	scale_features() output shape is (1, 7)	PASS
T06	predictor.py	predict_crop() rice inputs return 'rice'	PASS
T07	predictor.py	predict_crop() coffee inputs return 'coffee'	PASS
T08	predictor.py	predict_crop() confidence score between 0 and 100	PASS
T09	app.py	GET / returns 200 OK	PASS
T10	app.py	POST /predict with valid rice inputs returns result.html	PASS
T11	app.py	POST /predict with invalid pH shows flash error	PASS
T12	app.py	POST /predict logs entry to predictions.csv	PASS
T13	app.py	404 handler returns custom error page	PASS
T14	validate.js	Client-side JS blocks submit when N > 140	PASS

4. Response Time Benchmarking (JMeter 100 concurrent users)

Metric	Result	Target	Status
Average response time (GET /)	68 ms	< 500 ms	PASS
Average response time (POST /predict)	182 ms	< 2,000 ms	PASS
95th percentile response time	310 ms	< 2,000 ms	PASS
Throughput (requests/sec)	87 req/s	ε 50 req/s	PASS
Error rate (all endpoints)	0.0%	< 1%	PASS
Peak concurrent users handled	100	ε 100	PASS